

ELEC4631 Tutorial 3

1. Determine whether the function f is lpd, pd, lnd or lnsd around the given point x^* in the state-space S .

(a) $S = \mathbb{R}, x^* = 0, f(x) = x^2 - (1/25)x^4$.

(b) $S = \mathbb{R}, x^* = 0, f(x) = \begin{cases} 0, & x = 0 \\ 1 - \frac{\sin(x)}{x}, & x \neq 0 \end{cases}$

(c) $S = \mathbb{R}^3, x^* = (2, -1, 4), f(x, y, z) = -(x - 2)^2 - (y + 1)^4 - (2x - z)^2$.

(d) $S = \mathbb{R}^3, x^* = (2, -1, 4), f(x, y, z) = -(y + 1)^4 - (2x - z)^2$.

2. A dynamical system with state-space $S = \mathbb{R}^2$ satisfies the linear ODE

$$\dot{x}(t) = \begin{bmatrix} -3 & 3 \\ -2 & 2 \end{bmatrix} x(t),$$

with initial state $x(0) = x_0$. Consider the quadratic form $V : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$V(x) = 5x_1^2 - 14x_1x_2 + 10x_2^2.$$

- (a) Describe all equilibrium points of the system.
 - (b) Compute $\dot{V}(x) = \nabla V(x)^T f(x)$.
 - (c) Show that $V(x)$ is lpd around the origin.
 - (d) Show that $\dot{V}(x)$ is lnsd around the origin.
 - (e) Show that the origin is stable in the sense of Lyapunov.
3. Show that for the following ODEs the origin is a globally asymptotically stable equilibrium point using the suggested Lyapunov functions:

- (a)

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} -x + y - xy^2 \\ -x - y - x^2y \end{bmatrix}$$

$$V(x, y) = x^2 + y^2.$$

(b)

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} -x^3 + y^4 \\ -y^3(x+1) \end{bmatrix}$$

$$V(x, y) = x^2 + y^2.$$

(c)

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} -x + 4y \\ -x - y^3 \end{bmatrix}$$

$$V(x, y) = x^2 + 4y^2.$$

4. Consider again the RLC circuit from Lecture 4. It was shown that by using the circuit's total stored energy as a Lyapunov function we could show that $(0,0)$ is a globally asymptotically stable equilibrium point. However, a bit more argument was needed because $\dot{V}$ is lnsd rather than lnd around $(0,0)$. Now you will show that there is an alternative Lyapunov function such that $\dot{V}$ is lpd. Consider now a new candidate Lyapunov function of the form:

$$V(v_C, i) = \frac{1}{2}v_C^2 + \epsilon L v_C i + \frac{L}{2C}i^2,$$

where ϵ is some *positive* number that is to be determined. Note that V as defined above is a quadratic form.

- (a) Show that there is an $\epsilon > 0$ small enough such that V is a positive definite quadratic form and $\dot{V}$ is a negative definite quadratic form.
- (b) Show that $(0,0)$ is a globally asymptotically stable equilibrium point.
5. Complete the argument that the function $V(\theta, \omega)$

$$V(\theta, \omega) = \frac{1}{2}\omega^2 + \frac{1}{2}\left(\frac{1}{D}\omega + \theta\right)^2 + \frac{g}{r}\left(1 + \frac{1}{D^2}\right)(1 - \cos \theta)$$

from Lecture 6 is a Lyapunov function for the equilibrium point at $(0,0)$ of the damped pendulum by proving that:

- (a) $V(\theta, \omega) > 0$ for all $(\omega, \theta) \neq (0,0)$.
- (b) $\dot{V}(\theta, \omega) = -(D\omega^2 + \frac{g}{rD}\theta \sin \theta)$.
- (c) Suppose that $g(\theta)$ is a twice differentiable function of θ . From calculus we know that if $g'(\theta^*) = 0$ and $g''(\theta^*) > 0$ then the point θ^* is a local minimum in some interval $(\theta^* - r, \theta^* + r)$, for some $r > 0$. Use this fact to show that the function $g(\theta) = \theta \sin \theta$ is lpd in some open interval $(-r, r)$ for some $r > 0$.

6. Consider the mechanical mass-damper system shown in Fig. 1 that is made up of a unit mass attached to a nonlinear spring and a damper that has nonlinear dependence on the momentum of the mass. Only two coordinates are needed to

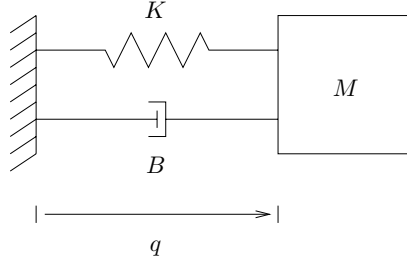


Figure 1: A mechanical mass-damper system

describe the motion of this system: the position $q(t)$ and the momentum $p(t) = \dot{q}(t)$. They satisfy the ODE:

$$\begin{aligned}\dot{q}(t) &= p(t), \\ \dot{p}(t) &= -f(p(t)) - g(q(t)),\end{aligned}$$

Here f and g are continuous nonlinear functions that model the friction in the damper and the restoring force of the spring, respectively. The functions f and g satisfy:

$$\begin{aligned}f(x) &= x^3 \quad \forall x \in [-\delta, \delta], \\ g(x) &= \sin(x)^5 \quad \forall x \in [-\delta, \delta],\end{aligned}$$

where δ is some given positive constant less than $\pi/2$. A candidate Lyapunov function for this mass-damper system is:

$$V(q, p) = \int_0^q g(u) du + \frac{1}{2}p^2.$$

Now,

- Show that $(0,0)$ is the only equilibrium point of the system in the region $-\delta \leq q \leq \delta$ and $-\delta \leq p \leq \delta$.
- Show that V is lpd around $(0,0)$.
- Compute $\dot{V}$ and show that it is lnsd around $(0,0)$. Therefore, V is a Lyapunov function and the system is stable i.s.L around $(0,0)$.
- Using the same Lyapunov function and the Krasovskii-LaSalle Invariance Principle, argue that in fact $(0,0)$ is an asymptotically stable equilibrium point.